

ACADEMIC DETAILS

| Degree    | Specialization            | Institute                                  | Year | CPI/% |
|-----------|---------------------------|--------------------------------------------|------|-------|
| M.Tech.   | Mechanical Engineering    | Indian Institute of Technology Gandhinagar | 2027 | 9.12  |
| B.Tech.   | Mechanical Engineering    | Chaitanya Bharathi Institute of Technology | 2024 | 9.48  |
| Class XII | Physics, Chemistry, Maths | Narayana Junior College                    | 2020 | 9.77  |
| Class X   |                           | Krishnaveni Talent School                  | 2018 | 9.5   |

INTERNSHIPS

- **Mechanical Intern – Thermo Fisher Scientific**

[May 2024 to Nov 2024 ]

(Medical Device Product Development | Value Engineering | Mechanical Design)

- Worked on cost reduction and value engineering of medical devices by redesigning mechanisms and eliminating unnecessary components through Proof of Concept (POC) development and validation.
  - Developed engineering concepts, performed mechanical calculations, created CAD models in SolidWorks, conducted cost and feasibility analysis, and supported prototype testing and validation.
- **Student Project Trainee – Bharat Heavy Electricals Limited (BHEL)**

[Aug 2023 to Sep 2023 ]

(Industrial Training | Thermal Power Plant Equipment | Manufacturing Operations)

- Studied the manufacturing processes and assembly of steam turbines, gas turbines, compressors, generators, and other critical equipment used in thermal power plants.
  - Gained practical exposure to thermal power plant operations, working cycles, industrial manufacturing practices, and production workflows through on-site training.

PROJECTS

- **Reduction of Emissions in a Single Cylinder Diesel Engine using Blends of Diesel & Cotton Seed Oil with Unconventional Catalytic Converter**

[Dec 2023 to April 2024]

(Prof. K.Kishor, CBIT) | [Project Link](#)

- Engineered and conducted a research study to optimize the blend of cottonseed oil and diesel fuel, achieved an 8% increase in mileage, an 8.7% enhancement in Brake Thermal Efficiency, and a 6.54% increment in air to-fuel ratio
  - Designed and fabricated a cost-effective catalytic converter utilizing Cerium oxide, charcoal, and Sponge Iron catalysts, resulting in a 58.8% reduction in CO emissions, a 15.21% decrease in HC emissions, and a significant 95.5% decrease in Hatridge Smoke Unit (HSU) values
- **Modeling and Control of a Soft Robotic Gripper with Variable Stiffness via Pneumatic and Tendon Actuation**

[March 2026 to May 2026]

(Prof. Madhu Vadali, IITGN) | [Project Link](#)

- Modeled a variable-stiffness soft robotic gripper using a nonlinear PRB model, derived its linearized state-space representation, and performed system analysis for controller design.
  - Designed and validated an LQR controller and Luenberger observer in MATLAB, achieving robust position and force regulation under disturbances and sensor noise through closed-loop simulations.
- **Techno-Economic and Policy Blueprint for the Ahmedabad–Gandhinagar Green Hydrogen Valley**

[March 2026 to May 2026]

(Prof. Atul Bhargav, IITGN) | [Project Link](#)

- Designed a 4 MW Green Hydrogen Valley by performing market demand assessment, electrolyzer selection, and sizing of hydrogen production, storage, and refuelling systems
  - Conducted techno-economic feasibility analysis using CAPEX/OPEX estimation, NPV and LCOH modelling, and policy evaluation to optimize project viability

TECHNICAL SKILLS

- SolidWorks, Python, C language, Auto Cad:

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant**

[Aug 2025 to April 2026]

- ES 115: Design, Innovation and Prototyping, ME 209 : Principles of Manufacturing Processes
  - Assisted in conducting laboratory sessions, guiding students in practical experiments and design activities. Evaluated assignments, addressed student queries, and supported course delivery under the supervision of the course instructor

ACHIEVEMENTS

- Placed in Dean’s List – IIT Gandhinagar (Semester I, AY 2025–26) for academic excellence.

• Secured 2nd position in Mechanical branch 2020-24 batch